

Reliability: What is it, and how is it measured?

Summary Therapists regularly perform various measurements. How reliable these measurements are in themselves, and how reliable therapists are in using them, is clearly essential knowledge to help clinicians decide whether or not a particular measurement is of any value. The aim of this paper is to explain the nature of reliability, and to describe some of the commonly used estimates that attempt to quantify it. An understanding of reliability, and how it is estimated, will help therapists to make sense of their own clinical findings, and to interpret published studies.

Although reliability is generally perceived as desirable, there is no firm definition as to the level of reliability required to reach clinical acceptability. As with hypothesis testing, statistically significant levels of reliability may not translate into clinically acceptable levels, so that some authors' claims about reliability may need to be interpreted with caution. Reliability is generally population specific, so that caution is also advised in making comparisons between studies.

The current consensus is that no single estimate is sufficient to provide the full picture about reliability, and that different types of estimate should be used together.

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Key Words

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Introduction

Therapists regularly perform various measurements of varying reliability. The term 'reliability' here refers to the consistency or repeatability of such measurements. Irrespective of the area in which they work, therapists take measurements for any or all of the reasons outlined in table 1. How reliable these measurements are in themselves, and how reliable therapists are in performing them, is

Table 1: Common reasons why therapists perform measurements

- As part of patient assessment.
- As baseline or outcome measures.
- As aids to deciding upon treatment plans.
- As feedback for patients and other interested parties.
- As aids to making predictive judgements, eg about outcome.

clearly essential knowledge to help clinicians decide whether or not a particular measurement is of any value.

This article focuses on the reliability of measures that generate *quantitative* data, and in particular 'interval' and 'ratio' data. Interval data have equal intervals between numbers but these are not related to true zero, so do not represent absolute quantity. Examples of interval data are IQ and degrees Centigrade or Fahrenheit. In the temperature scale, the difference between 10° and 20° is the same as between 70° and 80°, but is based on the numerical value of the scale, not the true nature of the variable itself. Therefore the actual difference in heat and molecular motion generated is not the same and it is not appropriate to say that someone is twice as hot as someone else.

With ratio data, numbers represent units with equal intervals, measured from true zero, eg distance, age, time, weight, strength, blood pressure, range of motion, height. Numbers therefore reflect actual amounts of the variable being measured, and it is appropriate to say that one person is twice as heavy, tall, etc, as another. The kind of quantitative measures that therapists often carry out are outlined in table 2.

The aim of this paper is to explain the nature of reliability, and to describe, in general terms, some of the commonly used methods for quantifying it. It is not intended to be a detailed account of the statistical

Table 2: Examples of quantitative measures performed by physiotherapists

- Strength measures (eg in newtons of force, kilos lifted).
- Angle or range of motion measures (eg in degrees, centimetres).
- Velocity or speed measures (eg in litres per minute for peak expiratory flow rate).
- Length or circumference measures (eg in metres, centimetres).

minutiae associated with reliability measures, for which readers are referred to standard books on medical statistics.

Measurement Error

It is very rare to find any clinical measurement that is perfectly reliable, as all instruments and observers or measurers (raters) are fallible to some extent and all humans respond with some inconsistency. Thus any observed score (X) can be thought of as a function of two components, ie a true score (T) and an error component (E): $X = T \pm E$

The difference between the true value and the observed value is measurement error. In statistical terms, 'error' refers to all sources of variability that cannot be explained by the independent (also known as the predictor, or explanatory) variable. Since the error components are generally unknown, it is only possible to estimate the amount of any measurement that is attributable to error and the amount that represents an accurate reading. This estimate is our measure of reliability.

Measurement errors may be systematic or random. Systematic errors are predictable errors, occurring in one direction only, constant and biased. For example, when using a measurement that is susceptible to a learning effect (eg strength testing), a retest may be consistently higher than a prior test (perhaps due to improved motor unit co-ordination). Such a systematic error would not therefore affect reliability, but would affect validity, as test values are not true representations of the quantity being measured. Random errors are due to chance and unpredictable, thus they are the basic concern of reliability.

Types of Reliability

Baumgartner (1989) has identified two types of reliability, ie relative reliability and absolute reliability.

Relative reliability is the degree to which individuals maintain their position in a sample over repeated measurements. Tables 3 and 4 give some maximum inspiratory pressure (MIP) measures taken on two occasions, 48 hours apart. In table 3, although the differences between the two measures vary from -16 to +22 centimetres of water, the ranking remains unchanged. That is, on both day 1 and day 2 subject 4 had the highest MIP, subject 1 the second highest, subject 5 the third highest, and so on. This form of reliability is often assessed

Table 3: Repeated maximum inspiratory pressure measures data demonstrating good relative reliability

Subject	MIP		Difference	Rank	
	Day 1	Day 2		Day 1	Day 2
1	110	120	+10	2	2
2	94	105	+11	4	4
3	86	70	-16	5	5
4	120	142	+22	1	1
5	107	107	0	3	3

Table 4: Repeated maximum inspiratory pressures measures data demonstrating poor relative reliability

Subject	MIP		Difference	Rank	
	Day 1	Day 2		Day 1	Day 2
1	110	95	-15	2	5
2	94	107	+13	4	3
3	86	97	+11	5	4
4	120	120	0	1	2
5	107	129	+22	3	1

by some type of correlation coefficient, eg Pearson's correlation coefficient, usually written as *r*. For table 3 the data give a Pearson's correlation coefficient of $r = 0.94$, generally accepted to indicate a high degree of correlation. In table 4, however, although the differences between the two measures look similar to those in table 1 (ie -15 to +22 cm of water), on this occasion the ranking has changed. Subject 4 has the highest MIP on day 1, but is second highest on day 2, subject 1 had the second highest MIP in day 1, but the lowest MIP on day 2, and so on. For table 4 data $r = 0.51$, which would be interpreted as a low degree of correlation. Correlation coefficients thus give information about *association* between two variables, and not necessarily about their proximity.

Absolute reliability is the degree to which repeated measurements vary for individuals, ie the less they vary, the higher the reliability. This type of reliability is expressed either in the actual units of measurement, or as a proportion of the measured values. The standard error of measurement (SEM), coefficient of variation (CV) and Bland and Altman's 95% limits of agreement (1986) are all examples of measures of absolute reliability. These will be described later.

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Why Estimate Reliability?

Reliability testing is usually performed to assess one of the following:

- Instrumental reliability, ie the reliability of the measurement device.
- Rater reliability, ie the reliability of the researcher/observer/clinician administering the measurement device.
- Response reliability, ie the reliability/stability of the variable being measured.

How is Reliability Measured?

As described earlier, observed scores consist of the true value $\pm$ the error component. Since it is not possible to know the true value, the true reliability of any test is not calculable. It can however be estimated, based on the statistical concept of variance, ie a measure of the variability of differences among scores within a sample. The greater the dispersion of scores, the larger the variance; the more homogeneous the scores, the smaller the variance.

If a single measurer (rater) were to record the oxygen saturation of an individual 10 times, the resulting scores would not all be identical, but would exhibit some variance. Some of this total variance is due to true differences between scores (since oxygen saturation fluctuates), but some can be attributable to measurement error (E). Reliability (R) is the measure of the amount of the total variance attributable to true differences and can be expressed as the ratio of true score variance (T) to total variance or:

$$R = \frac{T}{T + E}$$

This ratio gives a value known as a *reliability coefficient*. As the observed score approaches the true score, reliability increases, so that with zero error there is perfect reliability and a coefficient of 1, because the observed score is the same as the true score. Conversely, as error increases reliability diminishes, so that with maximal error there is no reliability and the coefficient approaches 0. There is, however, no such thing as a minimum acceptable level of reliability that can be applied to all measures, as this will vary depending on the use of the test.

Indices of Reliability

In common with medical literature, physiotherapy literature shows no consistency in authors' choice of reliability

estimate calculated for their data. Table 5 summarises the more common reliability indices found in the literature, which are described below.

Table 5: Reliability indices in common use

Hypothesis tests for bias, eg paired t-test, analysis of variance.
Correlation coefficients, eg Pearson's, ICC.
Standard error of measurement (SEM).
Coefficient of variation (CV).
Repeatability coefficient.
Bland and Altman 95% limits of agreement.

Indices Based on Hypothesis Testing for Bias

The paired t-test, and analysis of variance techniques are statistical methods for detecting systematic bias between groups of data. These estimates, based upon hypothesis testing, are often used in reliability studies. However, they give information only about systematic differences between the *means* of two sets of data, not about individual differences. Such tests should, therefore, not be used in isolation, but be complemented by other methods, eg Bland and Altman agreement tests (1986).

Correlation Coefficients (r)

As stated earlier, correlation coefficients give information about the degree of association between two sets of data, or the consistency of position within the two distributions. Provided the relative positions of each subject remain the same from test to test, high measures of correlation will be obtained. However, a correlation coefficient will not detect any systematic errors. So it is possible to have two sets of scores that are highly correlated, but not highly repeatable, as in table 6 where the hypothetical data give a Pearson's correlation coefficient of $r = 1$, ie perfect correlation despite a systematic difference of 40 cm of water for each subject.

Thus correlation only tells how two sets of scores vary together, not the extent of agreement between them. Often researchers need to know that the actual values obtained by two measurements are the same, not just proportional to one another. Although published studies abound with correlation used as the sole indicator of reliability, their results can be misleading, and it is now recommended that they be no longer used in isolation (Keating and Matyas, 1998; Chinn, 1990).

Table 6: Repeated maximum inspiratory pressures measures data demonstrating a high Pearson's correlation coefficient, but poor absolute reliability

Subject	MIP		Difference	Rank	
	Day 1	Day 2		Day 1	Day 2
1	110	150	+40	2	2
2	94	134	+40	4	4
3	86	126	+40	5	5
4	120	160	+40	1	1
5	107	147	+40	3	3

Intra-class Correlation Coefficient (ICC)

The intra-class correlation coefficient (ICC) is an attempt to overcome some of the limitations of the classic correlation coefficients. It is a single index calculated using variance estimates obtained through the partitioning of total variance into between and within subject variance (known as analysis of variance or ANOVA). It thus reflects both degree of consistency and agreement among ratings.

There are numerous versions of the ICC (Shrout and Fleiss, 1979) with each form being appropriate to specific situations. Readers interested in using the ICC can find worked examples relevant to rehabilitation in various published articles (Rankin and Stokes, 1998; Keating and Matyas, 1998; Stratford *et al*, 1984; Eliasziw *et al*, 1994). The use of the ICC implies that each component of variance has been estimated appropriately from sufficient data (at least 25 degrees of freedom), and from a sample representing the population to which the results will be applied (Chinn, 1991). In this instance, degrees of freedom can be thought of as the number of subjects multiplied by the number of measurements.

As with other reliability coefficients, there is no standard acceptable level of reliability using the ICC. It will range from 0 to 1, with values closer to one representing the higher reliability. Chinn (1991) recommends that any measure should have an intra-class correlation coefficient of at least 0.6 to be useful. The ICC is useful when comparing the repeatability of measures using different units, as it is a dimensionless statistic. It is most useful when three or more sets of observations are taken, either from a single sample or from independent samples. It does, however, have some disadvantages as described by Rankin and Stokes (1998) that make it unsuitable for use in isolation. As described earlier, any reliability coefficient is determined as the ratio of variance between

subjects to the sum of error variance and subject variance. If the variance between subjects is sufficiently high (that is, the data come from a heterogeneous sample) then reliability will inevitably appear to be high. Thus if the ICC is applied to data from a group of individuals demonstrating a wide range of the measured characteristic, reliability will appear to be higher than when applied to a group demonstrating a narrow range of the same characteristic.

Standard Error of Measurement (SEM)

As mentioned earlier, if any measurement test were to be applied to a single subject an infinite number of times, it would be expected to generate responses that vary a little from trial to trial, as a result of measurement error. Theoretically these responses could be plotted and their distribution would follow a normal curve, with the mean equal to the true score, and errors occurring above and below the mean.

The more reliable the measurement response, the less error variability there would be around the mean. The standard deviation of measurement errors is therefore a reflection of the reliability of the test response, and is known as the standard error of measurement (SEM). The value for the SEM will vary from subject to subject, but there are equations for calculating a group estimate, eg $SEM = s_x \sqrt{1 - r_{xx}}$ (where s_x is the standard deviation of the set of observed test scores and r_{xx} is the reliability coefficient for those data – often the ICC is used here.)

The SEM is a measure of absolute reliability and is expressed in the actual units of measurement, making it easy to interpret, ie the smaller the SEM, the greater the reliability. It is only appropriate, however, for use with interval data (Atkinson and Neville, 1998) since with ratio data the amount of random error may increase as the measured values increase.

Coefficient of Variation (CV)

The CV is an often-quoted estimate of measurement error, particularly in laboratory studies where multiple repeated tests are standard procedure. One form of the CV is calculated as the standard deviation of the data, divided by the mean and multiplied by 100 to give a percentage score. This expresses the standard deviation as a proportion of the mean, making it unit independent. However, as Bland (1987) points out, the problem with expressing the error as a percentage, is that $x\%$ of the smallest observation will differ markedly from $x\%$ of the largest observation. Chinn (1991) suggests that it is preferable to use the ICC rather than the CV, as the former relates the size of the error variation to the size of the variation of interest. It has been suggested that the above form of the CV should no longer be used to estimate reliability, and that other more appropriate methods should be employed based on analysis of variance of logarithmically transformed data (Atkinson and Neville, 1998).

Repeatability Coefficient

Another way to present measurement error over two tests, as recommended by the British Standards Institution (1979) is the value below which the difference between the two measurements will lie with probability 0.95. This is based upon the within-subject standard deviation (s). Provided the measurement errors are from a normal distribution this can be estimated by $1.96 \times \sqrt{(2s^2)}$, or $2.83s$ and is known as the repeatability coefficient (Bland and Altman, 1986). This name is rather confusing, as other coefficients (eg reliability coefficient) are expected to be unit free and in a range from zero to one. The method of calculation varies slightly in two different references (Bland and Altman, 1986; Bland, 1987), and to date it is not a frequently quoted statistic.

Bland and Altman Agreement Tests

In 1986 *The Lancet* published a paper by Bland and Altman that is frequently cited and has been instrumental in encouraging changing use of reliability estimates in the medical literature. In the past, studies comparing the reliability of two different instruments designed to measure the same variable (eg two different types of goniometer) often quoted correlation coefficients and ICCs. These can both be misleading, however, and are not

appropriate for method comparison studies for reasons described by Bland and Altman in their 1986 paper. These authors have therefore proposed an approach for assessing agreement between two different methods of clinical measurement. This involves calculating the mean for each method and using this in a series of agreement tests.

Step 1 consists of plotting the difference in the two results against the mean value from the two methods. Step 2 involves calculating the mean and standard deviation of the *differences* between the measures. Step 3 consists of calculating the 95% limits of agreement (as the mean difference plus or minus two standard deviations of the differences), and 95% confidence intervals for these limits of agreement. The advantages of this approach are that by using scatterplots, data can be visually interpreted fairly swiftly. Any outliers, bias, or relationship between variance in measures and size of the mean can therefore be observed easily. The 95% limits of agreement provide a range of error that may relate to clinical acceptability, although this needs to be interpreted with reference to the range of measures in the raw data.

In the same paper, Bland and Altman have a section headed 'Repeatability' in which they recommend the use of the 'repeatability coefficient' (described earlier) for studies involving repeated measures with the same instrument. In their final discussion, however, they suggest that their agreement testing approach may be used either for analysis of repeatability of a single measurement method, or for method comparison studies. Worked examples using Bland and Altman agreement tests can be found in their original paper, and more recently in papers by Atkinson and Nevill (1998) and Rankin and Stokes (1998).

Nature of Reliability

Unfortunately, the concept of reliability is complex, with less of the straightforward 'black and white' statistical theory that surrounds hypothesis testing. When testing a research hypothesis there are clear guidelines to help researchers and clinicians decide whether results indicate that the hypothesis can be supported or not. In contrast, the decision as to whether a particular measurement tool or method is reliable or not is more open to interpretation. The decision to be made is whether the level of measurement error is

considered acceptable for practical use. There are no firm rules for making this decision, which will inevitably be context based. An error of $\pm 5^\circ$ in goniometry measures may be clinically acceptable in some circumstances, but may be less acceptable if definitive clinical decisions (eg surgical intervention) are dependent on the measure. Because of this dependence on the context in which they are produced, it is therefore very difficult to make comparisons of reliability across different studies, except in very general terms.

Conclusion

This paper has attempted to explain the concept of reliability and describe some of the estimates commonly used to quantify it. Key points to note about reliability are summarised in the panel below. Reliability should not necessarily be conceived as a property that a particular instrument or measurer does or does not possess. Any

instrument will have a certain degree of reliability when applied to certain populations under certain conditions. The issue to be addressed is what level of reliability is considered to be clinically acceptable. In some circumstances there may be a choice only between a measure with lower reliability or no measure at all, in which case the less than perfect measure may still add useful information.

In recent years several authors have recommended that no single reliability estimate should be used for reliability studies. Opinion is divided over exactly which estimates are suitable for which circumstances. Rankin and Stokes (1998) have recently suggested that a consensus needs to be reached to establish which tests should be adopted universally. In general, however, it is suggested that no single estimate is universally appropriate, and that a combination of approaches is more likely to give a true picture of reliability.

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Key Messages

Reliability is:

- Not an all-or-none phenomenon.
- Open to interpretation.
- Not the same as clinical acceptability.
- Population specific.
- Related to the variability in the group studied.
- Best estimated by more than one index.